INCH-POUND

MIL-R-83579

6 April 1994

MILITARY SPECIFICATION

RESISTORS, FIXED, WIRE-WOUND (POWER TYPE), HIGH RELIABILITY, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for axial lead precision, power type, wire-wound, fixed resistors having a 25°C ambient operating temperature derated to zero load at 250°C for use in electrical, electronic communication, and other associated equipment and for space, missile and other high reliability applications. The prerequisite to qualifying to this specification shall be based on current qualification to "S" failure rate level for each detail specification of MIL-R-39007. Resistors of this specification shall be capable of meeting all requirements of MIL-R-39007 in addition to the requirements specified herein. These resistors have initial resistance tolerances of ±.5, ±.1, and ±1 percent as shown in table IV. Resistors which are noninductively wound with solderable leads are designated "N" while weldable leads are designated "Z". Resistors designated by the letter "S" and "U" are inductively wound with solderable and weldable leads, respectively (see 1.2.1.2).

1.2 Classification.

1.2.1 Part or Identifying Number (PIN). The PIN shall be in the following form, and as specified (see 3.1 and 6.1):

M83579XX	S	49R9	F
Specification sheet (1.2.1.1)	Resistor winding and lead type (1.2.1.2)	Resistance (1.2.1.3)	Resistance tolerance (1.2.1.4)

1.2.1.1 Style. The style is identified by the three-letter symbol "RWS" which represents high-reliability, axial-lead, power type, wire-wound, fixed resistors, and is to be used for reference only.

1.2.1.2 Resistor winding and lead type. The resistor winding and the resistor lead type shall be as defined in table I.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: SMC/SDFP, P.O. Box 92960, Worldway Postal Center, Los Angeles, CA 90009-2960, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of the document or by letter.

AMSC N/A FSC 5905

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:46Z
Check the source to verify that this is the current version before use.

TABLE I. Resistor winding and resistor lead type.

Type	Resistor winding and lead type
S	Solderable, inductively wound
W	Weldable, inductively wound
N	Solderable, noninductively wound 1/
Z	Weldable, noninductively wound 1/

1/ Wound using Ayrton-Perry method.

1.2.1.3 Resistance. The nominal resistance is expressed in ohms and is identified by four digits; the first three digits represent significant figures and the last digit specifies the number of zeros to follow. When the value of resistance is less than 100 ohms, or when fractional values of an ohm are required, the letter "R" shall be substituted for one of the significant digits to represent the decimal point. When the letter "R" is used, succeeding digits of the group represent significant figures. The resistance value designations are shown in table II. Minimum and maximum resistance values shall be as specified (see 3.1). The standard resistance values for every decade shall follow the sequence demonstrated for 10 to 100 decade in table III. For resistance tolerances F and D, only those resistance values which follow the sequence of values listed in the 10 to 100 decade in table III shall be considered as conforming to this specification.

TABLE II. Resistance values.

Designation	Resistance (ohms)
R100 to R988 inclusive	0.100 to 0.988 inclusive
1R00 to 9R88 inclusive	1.00 to 9.88 inclusive
10R0 to 98R8 inclusive	10.0 to 98.8 inclusive
1000 to 9880 inclusive	100 to 988 inclusive
1001 to 9881 inclusive	1,000 to 9,880 inclusive
1002 to 4022 inclusive	10,000 to 40,200 inclusive

TABLE III. Standard resistance values for the 10 to 100 decade.

0.1 (B) 1/	0.5 (D)	1.0 (F)
10.00		10.00
10.10		.....
10.20		10.20
10.40		.....
10.50		10.50
10.60		.....
10.70		10.70
10.90		.....
11.00		11.00
11.10		.....
11.30		11.30
11.40		.....
11.50		11.50
11.70		.....
11.80		11.80
12.00		.....
12.10		12.10
12.30		.....
12.40		12.40
12.60		.....
22.90		.....
23.20		23.20
23.40		.....
23.70		23.70
24.00		.....
24.30		24.30
24.60		.....
24.90		24.90
25.20		.....
25.50		25.50
25.80		.....
26.10		26.10
26.40		.....
26.70		26.70
.....		.....
27.10		.....
27.40		27.40
27.70		.....
28.00		28.00
28.40		.....
49.30		.....
49.90		49.90
50.50		.....
.....		.....
51.10		51.10
51.70		.....
52.30		52.30
53.00		.....
53.60		53.60
54.20		.....
54.90		54.90
55.60		.....
.....		.....
56.20		56.20
56.90		.....
57.60		57.60
58.30		.....
59.00		59.00
59.70		.....
60.40		60.40

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:46Z
Check the source to verify that this is the current version before use.

TABLE III. Standard resistance values for the 10 to 100 decade - Continued.

0.1 (B) 1/	0.5 (D)	1.0 (F)
12.70		12.70
12.90		.....
13.00		13.00
13.20		.....
13.30		13.30
13.50		.....
13.70		13.70
13.80		.....
14.00		14.00
14.20		.....
14.30		14.30
14.50		.....
14.70		14.70
14.90		.....
15.00		15.00
15.20		.....
15.40		15.40
15.60		.....
15.80		15.80
16.00		.....
16.20		16.20
16.40		.....
16.50		16.50
16.70		.....
16.90		16.90
17.20		.....
17.40		17.40
17.60		.....
17.80		17.80
18.00		18.00
18.20		18.20
18.40		.....
18.70		18.70
18.90		.....
19.10		19.10
19.30		.....
19.60		19.60
19.80		.....
20.00		20.00
20.30		.....
20.50		20.50
20.80		.....
21.00		21.00
21.30		.....
21.50		21.50
21.80		.....
.....		.....
22.10		22.10
22.30		.....
22.60		22.60
28.70		28.70
29.10		.....
29.40		29.40
29.80		.....
.....		.....
30.10		30.10
30.50		.....
30.90		30.90
31.20		.....
31.60		31.60
32.00		.....
32.40		32.40
32.80		.....
.....		.....
33.20		33.20
33.60		.....
34.00		34.00
34.40		.....
34.80		34.80
35.20		.....
35.70		35.70
.....		.....
36.10		.....
36.50		36.50
37.00		.....
37.40		37.40
37.90		.....
38.30		38.30
38.80		.....
.....		.....
39.20		39.20
39.70		.....
40.20		40.20
40.70		.....
41.20		41.20
41.70		.....
42.20		42.20
42.70		.....
.....		.....
43.20		43.20
43.70		.....
44.20		44.20
44.80		.....
45.30		45.30
45.90		.....
46.40		46.40
47.00		.....
47.50		47.50
48.10		.....
48.70		48.70
61.20		.....
61.90		61.90
.....		.....
62.60		.....
63.40		63.40
64.20		.....
64.90		64.90
65.70		.....
66.50		66.50
67.30		.....
.....		.....
68.10		68.10
69.00		.....
69.80		69.80
70.60		.....
71.50		71.50
72.30		.....
73.20		73.20
74.10		.....
75.00		75.00
75.90		.....
76.80		76.80
77.70		.....
78.70		78.70
79.60		.....
80.60		80.60
81.60		.....
.....		.....
82.50		82.50
83.50		.....
84.50		84.50
85.60		.....
86.60		86.60
87.60		.....
88.70		88.70
89.80		.....
90.90		90.90
.....		.....
92.00		.....
93.10		93.10
94.20		.....
95.30		95.30
96.50		.....
97.60		97.60
98.80		.....

1/ The resistance values for tolerance B may be of any value, but it is preferred that the values be chosen from the tolerance D values in this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:46Z
Check the source to verify that this is the current version before use.

1.2.1.4 Resistance tolerance. The resistance tolerance is identified by a single letter in accordance with table IV.

TABLE IV. Resistance tolerance.

Symbol	Resistance tolerance
	Percent (±)
B	0.1
D	0.5
F	1.0

1.2.1.4.1 Replacement or maintenance (applicable to 0.1 and 0.5 percent tolerances). It is preferred that resistance values be selected as specified in table III. However, if it is impracticable to do so, then any value (within specification limits) may be specified and these values shall be considered as conforming to this specification (see table V and 3.1).

TABLE V. Designation of resistance values (0.1 and 0.5 percent tolerances).

Designation	Resistance value
R100 to R999 inclusive	.100 to .999 inclusive
1R00 to 9R99 inclusive	1.00 to 9.99 inclusive
10R0 to 99R9 inclusive	10.0 to 99.9 inclusive
1000 to 9990 inclusive	100 to 999 inclusive
1001 to 9991 inclusive	1,000 to 9,990 inclusive
1002 to maximum	10,000 to maximum

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-14256 - Flux.
MIL-R-39007 - Resistors, Fixed, Wire-Wound (Power Type), Established Reliability, General Specification For.
MIL-R-39032 - Resistors, Packaging of.
MIL-R-83579/1 - Resistor, Fixed, Wire-wound (Power Type), Established Reliability, Style RWS81.
MIL-R-83579/2 - Resistor, fixed, Wire-wound (Power Type), Established Reliability, Style RWS84.
MIL-R-83579/3 - Resistor, Fixed, Wire-wound (Power Type), Established Reliability, Style RWS89.

STANDARDS

MILITARY

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-790 - Product Assurance Program for Electronic Parts Specification.
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines
MIL-STD-1276 - Leads, Weldable, for Electronic Component Parts.
MIL-STD-1285 - Marking of Electrical and Electronic Parts.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:46Z
Check the source to verify that this is the current version before use.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION (NASA)

NASA 1124 - Outgassing Data for Selecting Spacecraft Materials.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 40, 700 Robbins Avenue, Philadelphia, PA 19111-509.)

2.2 Non-Government publications. The following document form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E595 - Standard Test Method for Total Mass Loss and Collected Volatile Condensable Materials from Outgassing in a Vacuum Environment.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and references cited herein (except for associated detail specifications, specifications sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specifications. Associated detail requirements or exceptions applicable to individual types of resistors shall be as specified in the associated detail specifications listed in this specification. In the event of any conflict between requirements of this specification and the specifications sheet, the latter shall govern (see 6.1).

3.2 Qualification. Resistors furnished under this specification shall be products which are qualified for listing on both qualified products lists for MIL-R-39007 (with minimum failure rate 'S') and MIL-R-83579. The manufacturer shall obtain certification from the qualifying activity that the reliability assurance requirements of 4.1.3 have been met and are being maintained. Distributors should not be used for distribution of parts. In addition, as a prerequisite for qualification, the manufacturer shall demonstrate to the qualifying activity that the requirements for outgassing have been met (see 3.23).

3.2.1 Reliability and baseline documentation. Resistors supplied to this specification shall conform to the requirements of MIL-STD-790 and the requirements specified herein. Prior to granting of qualification approval, a baseline document shall be prepared by the manufacturer and submitted to the qualifying activity for approval (see appendix A).

3.3 Material. The material shall be as specified herein and in accordance with the baseline document (see 3.2.1). In addition, materials used in the manufacturing of the resistor shall meet the outgassing requirements (see 3.23).

3.4. Terminal leads. Terminal leads shall be made of a solid conductor of the length and diameter specified. The lead composition and final finish shall be specified by C32 or C52 (maximum thickness of 200 microinches is not applicable), W32 or W52 per MIL-STD-1276. In addition, manufacturers may supply type N22 or N01 lead finish. The leads shall be solderable per MIL-STD-202, method 208, beyond the maximum specified clean lead-to-clean lead dimension.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:46Z
Check the source to verify that this is the current version before use.

3.4.1 Solderable terminals. Solderable leads shall be suitably treated to meet the requirements of solderability (see 4.7.4). At the option of the manufacturer, the terminals may be solder coated or otherwise treated to meet solderability requirements following the conditioning test specified in 4.7.2. When a lead coating containing tin is used, the tin content shall range between 40 and 70 percent (see 3.1).

3.4.2 Weldable terminals. Weldable terminals shall be type N1 of MIL-STD-1276. The manufacturer shall verify by certification that the weldable terminals meet all the requirements for type N1 of MIL-STD-1276 (see 3.1). The solderability requirement of 3.10 is not applicable to weldable terminals.

3.4.3 Flux. The manufacturer shall verify by certification that only noncorrosive solder fluxes shall be used unless it can be shown that corrosive elements have been removed satisfactorily after soldering. Noncorrosive fluxes shall comply with the requirements for resistivity of water extract and effect on vacuum-deposited copper specified in MIL-F-14256. If cored solders are used, the proportion of flux to solder by volume shall be between 1 percent and 3 percent.

3.4.4 End caps. When end caps are used in construction of the resistor, them is misalignment of the cap with respect to the core shall not exceed 5 degrees.

3.4.5 Insulation. Insulation material surrounding the resistor body shall extend around the entire shoulder of each end cap toward the axis of the lead attachments and cover the internal resistive body element.

3.4.6 Solder dip (retinning) leads. The manufacture may solder dip/retin the leads of product supplied to this specification provided the solder dip process has been approved by the qualifying activity.

3.4.6.1 Qualifying activity approval. Approval of the solder dip process shall be based on one of the following options:

a. When the original lead finish qualified was hot solder dip lead finish 52 of MIL-STD-1276 (Note: The 200 microinch maximum thickness is not applicable). The manufacture shall use the same solder dip process for retinning as is used in the original manufacture of the product.

b. When the lead originally qualified was not hot solder dip lead finish 52 of MIL-STD-1276 as prescribed in 3.4.6.1a, approval for the process to be used for solder dip shall be based on the following test procedure:

(1) Thirty samples of any resistance value for each style and lead finish are subjected to the manufacturer's solder dip process. (Note: If hermetic and nonhermetic constructions are available, 30 samples of the hermetic and 30 samples of the nonhermetic construction shall be tested. Also, the seal test shall be performed in place of the moisture resistance test for hermetic construction.) Following the solder dip process, the resistors are subjected to the direct current resistance test and other group A electricals. (Note: If hermetic seal testing is required in group A, these tests would also be performed.) No defects are allowed.

(2) Ten of the 30 samples are then subjected to the solderability test. No defects are allowed.

(3) The remaining 20 samples are subjected to the resistance to solder heat followed by the moisture resistance test. No defects are allowed.

(Note: Solder dip of gold leads is not allowed.)

3.4.6.2 Solder dip/retinning options. The manufacture may solder dip/retin as allows:

a. After the 100 percent group A screening test and following the solder dip/retinning process, the electrical measurements required in group A, subgroup 1, 100 percent screening test shall be repeated on 100 percent of the lot (Note: The manufacturer may dip/retin prior to the 100 percent electrical measurements of the group A, subgroup 1 tests). The Percent Defective Allowable (PDA) for the electrical measurements shall be as for the subgroup 1, tests.

b. As a corrective action if the lot fails the group A solderability test.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:46Z
Check the source to verify that this is the current version before use.

c. After the group A inspection has been completed and following the solder dip/retinning process, the electrical measurements required in group A, subgroup 1, 100 percent screening test shall be repeated on 100 percent of the lot. The PDA for the electrical measurements shall be as for the subgroup 1 tests. Following this test, the manufacture shall submit the lot to the group A solderability test as specified in 4.7.4.

3.4.7 Tin plated finishes. Use of tin plating is prohibited as a final finish and as an undercoat (see 6.6.1). Use of tin-lead (Sn-Pb) finishes are acceptable provided that the minimum lead content is 3 percent.

3.5 Design and construction. Resistors shall be of the design, construction, and physical dimensions specified (see 3.1). The design of the resistors shall be such as to preclude shorting of turns and to obtain a minimum voltage drop between adjacent turns. Except as required for noninductive performance capability, resistors shall be wound with a single layer of resistance wire. For terminal wire and winding symbols "N" and "Z", resistors shall be wound by the Ayrton-Perry method.

3.5.1 Windings.

3.5.1.1 Wire. The manufacturer shall verify by certification that the wire has no joints, welds, or bonds, except at the end terminals where welding, brazing, or silver solder only shall be employed. The wire shall possess a uniform cross section of conductor and insulation (if employed), and shall be as free as practicable from particles or impurities, grain growth, or other factors contributing to stop weakness. The cross sectional area of the wire shall be the maximum consistent with other requirements specified herein. Where abrasive resistance value adjustment is used, the abrasion must be evenly distributed around the resistor body. The remaining cross sectional area of the wire, after the abrading operation, shall be at least 90 percent of the original cross sectional area.

3.5.1.2 Pitch. The average winding pitch shall not exceed five times the wire diameter.

3.5.1.3 Effective wire coverage. Effective wire coverage is the winding length on the tube between points of departure from the nominal winding pitch. The effective wire coverage shall be such that not more than 20 percent of the overall winding area shall remain uncovered (by wire).

3.5.1.4 Protective coating or enclosure. Resistor assemblies shall be protected by a coating or moisture resistant insulating material which shall completely cover the exterior of the resistance element, including connections or terminations. The coating shall not crack, craze, run, or form globules at any temperature up to and including 250°C, regardless of the mounting position of the resistor. This material shall afford adequate protection against the effects of prolonged exposure to high humidities. The protective coating or enclosure shall be such as to minimize the establishment of leakage paths between the terminals resulting from collection of moisture film on the exterior surface of the resistor.

3.5.1.5 Weight. Resistors shall not exceed the maximum weight specified (see 3.1).

3.6 Power rating. Resistors shall have a power rating based on continuous full load operation at an ambient temperature of 25°C (see 3.1). This power rating is dependent on the ability of resistors to meet the S level failure rate requirements specified in MIL-R-39007. For temperatures in excess of those specified above, the load shall be derated in accordance with figure 1.

3.7 Voltage rating. Resistors shall have a rated direct current (dc) continuous working voltage, or an approximate sine wave root mean square (rms) continuous working voltage at commercial line frequency and waveform corresponding to the power rating, as determined from the following formula:

$$E=\sqrt{(PR)}$$

Where; E = continuous rated dc or rms working voltage in volts
P = rated wattage in watts
R = nominal resistance in ohms

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:46Z
Check the source to verify that this is the current version before use.

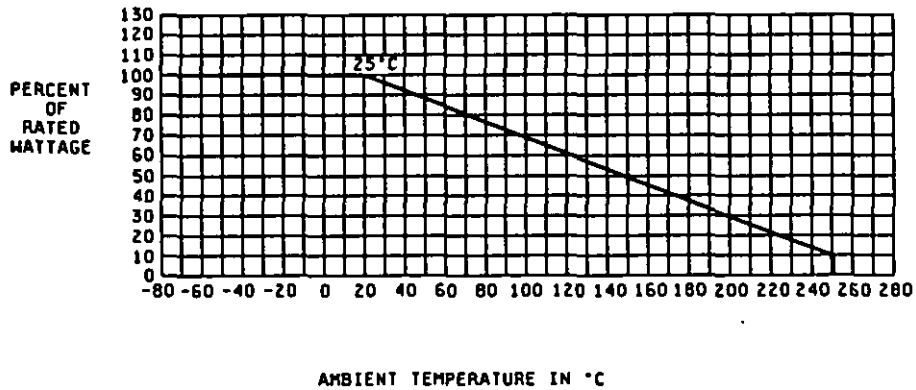


FIGURE 1. Derating curve for high ambient temperatures.

3.8 Conditioning. When resistors are tested as specified in 4.7.2, there shall be no mechanical damage. The change in resistance shall not exceed ±(0.2 percent +.005 ohm).

3.9 DC resistance. When resistors are tested as specified in 4.7.3, the dc resistance shall be within the specified tolerance of the nominal resistance (see 1.1 and 3.1).

3.10 Solderability (applicable to S and N terminals only). When resistors are tested as specified in 4.7.4, they shall meet the criteria for wire lead terminal evaluation in the test method.

3.11 Resistance to solvents. When resistors are tested as specified in 4.7.5, there shall be no evidence of mechanical damage and marking shall remain legible.

3.12 Thermal shock. When resistors are tested as specified in 4.7.6, there shall be no evidence of mechanical damage; the change in resistance shall not exceed ±(0.2 percent +.005 ohm).

3.13 Short-time overload. When resistors are tested as specified in 4.7.8, there shall be no evidence of arcing, burning, or charring; the change in resistance shall not exceed ±(0.2 percent +.005 ohm).

3.14 Dielectric withstanding voltage. When resistors are tested as specified in 4.7.9, the leakage rate shall not exceed 500 microamperes; there shall be no evidence of flashover, mechanical damage, arcing or insulation breakdown; the change in resistance shall not exceed ±(0.1 percent +.005 ohm).

3.15 Insulation resistance. When resistors are tested as specified in 4.7.10, the insulation resistance shall not be less than 1,000 megohms.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:46Z
Check the source to verify that this is the current version before use.

3.16 Moisture resistance. When resistors are tested as specified in 4.7.11, there shall be no evidence mechanical damage; the change in resistance shall not exceed ±(0.2 percent +.005 ohm). In addition, the dielectric withstanding voltage shall be as specified in 3.14, and the insulation resistance shall be 100 megohms, minimum.

3.17 Maximum pulse voltage. The maximum pulse voltage shall be as specified (see 3.1).

3.17.1 Limitations.

a. Under reduced pressure conditions, the voltage shall not exceed the values specified (see 3.1) due to reduced dielectric strength of the air.

b. When the resistors are operated at temperatures above +25°C, the pulse power rating shall be drafted (see 3.1).

c. When the resistors are operated under steady-state conditions and a pulse is applied in addition, the pulse power raring shall be drafted so that the sum of the steady-state power pulse and the pulse power does not exceed derating (see 3.1).

3.18 Destructive physical analysis (DPA). When examined as specified in 4.7.12 resistors shall meet the requirements in appendix B of this document.

3.19 Outgassing. When examined as specified in 4.7.13, the samples shall meet the following requirements:

a. Total Mass Loss (TML) shall not exceed 1 percent.

b. Volatile Condensable Material (VCM) shall not exceed .1 percent.

Data listed in NASA Publication 1124, may be in lieu of actual test data for applicable materials.

3.20 Fungus. All external materials shall be nonnutrient to fungus growth or shall be suitably treated to retard fungus growth. The manufacturer shall verify by certification that all external materials are fungus resistant or shall test the resistors as specified in 4.7.14. There shall be no evidence of fungus growth on the external surfaces.

3.21 Reactance (applicable to "N" and "Z" terminals and windings only). When resistors are tested a specified in 4.7.15, they shall be within the maximum limits specified in table VI.

TABLE VI. Reactance.

Styles	Maximum effective series inductance μH		Maximum effective parallel capacitance - pF
	Below 50 ohms	above 50 ohms	All resistance values
RWS81	0.20	0.37	1.5
RWS84	0.30	0.60	1.5
RWS89	0.20	0.37	1.5

3.22 Radiographic inspection. When resistors are tested in accordance with this test, they shall exhibit no evidence of defective welds (see 4.7.16). All testing shall be done in accordance with appendix C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:46Z
Check the source to verify that this is the current version before use.

3.23 Marking. Resistors shall be marked with the type designation, "JAN" marking, date code, source code, and manufacturer's production lot code (see MIL-STD-790). Date and source code shall be in accordance with MIL-STD-1285. At the option of the manufacturer, the type designation shall be divided between the characteristic letter and the first digit of the resistance value. The following is an example of complete marking:

12345	-	Source code
6633J	-	Date code and "JAN" marking
M83579XX	-	PIN
XXXXXX	-	Resistor winding and lead type, Resistance, Resistance tolerance.
XXXXX	-	Manufacturer's production lot code (any letter except "J")

The date code shall provide traceability through all production operations and shall represent the date of winding or enclosure as consistently provided by the manufacturer.

3.23.1 JAN and J marking. The United States Government has adopted, and is exercising legitimate control over, the certification marks "JAN and J", respectively , to indicate that items so marked or identified are manufactured to, and meet all the requirements of military specifications. Accordingly, items acquired to, and meeting all of the criteria specified herein and in applicable detail specifications shall bear the certification mark "JAN", except that items too small to bear the certification mark "JAN" shall bear the letter "J". The "JAN" or "J" shall be placed immediately before the part number except that if such location would place a hardship on the manufacturer in connection with such marking, the "JAN" or"J" may be located on the first line above or below the part number. Items furnished under contract or orders which either permit or require deviation from the conditions or requirements specified herein or in applicable specifications shall not bear "JAN" or "J". In the event a item sample fails to meet the requirements of this specification and applicable detail specification, the manufacturer shall remove the "JAN" or the "J" from the sample tested and also from all items represented by sample. The "JAN" and "J" certification mark shall not be used on products acquired to contractor drawings or specifications. The United States Government has obtained Certificate of Registration No. 504,860 for the Certification mark "JAN" or "J".

3.23.2 Minimum marking. When the physical size of the resistor style precludes the marking of all of the information cited in the example in 3.26, the minimum marking required shall be as specified in the detail specification (see 3.1). Marking shall remain legible at the end of all tests. In those cases where full marking requirements are not on the resistor body, full marking shall be marked on the unit package.

3.23.3 Use of conductive inks. Conductive inks shall not be used to coat the body of the resistor as preparation for marking or used for marking of resistors.

3.23.4 Beryllium oxide (BeO). Manufacturers which use beryllium oxide in their construction shall mark each resistor body and resistor package with the symbol "BeO".

3.24 Supplying to higher resistance tolerances. Parts qualified and marked to lower tolerances, with procuring agency approval, are substitutable for parts marked to higher tolerance levels and shall not be remarked unless specified in the contract or purchase order (see 6.1)(see table VII).

TABLE VII. Resistance tolerance substitution.

Resistance tolerance	Resistance tolerance substitution
B	
D	B
F	B, D

3.25 Workmanship. Resistors shall be processed in such a manner as to be uniform in quality and shall meet the requirements of 3.4 to 3.5.1.6, inclusive, and shall be free from defects that will affect, serviceability, or appearance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:46Z
Check the source to verify that this is the current version before use.

3.25.1 Soldering. Where soldering is employed, only noncorrosive fluxes shall be used, unless it can be shown that corrosive elements have been satisfactorily removed after soldering. Solder shall not be used for obtaining mechanical strength. Electrical connections shall be mechanically secure before soldering and electrically continuous after soldering. Except for solder used to coat the terminals, the solder used shall in no case start to melt at a temperature less than 300°C.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality and quantity to, which permit performance of the required inspection shall be established and maintained by the supplier. The establishment and maintenance of a calibration systems to control the accuracy of the measuring and test equipment shall be in accordance with MIL-STD-45662.

4.1.3 Product assurance program. A product assurance program shall be established and maintained in accordance with MIL-STD-790. Evidence of such compliance shall be verified by the qualifying activity of this specification as a prerequisite for qualification and continued qualification.

4.2 Classification of inspection. The examination and testing of resistors shall be classified as follows:

a. Qualification inspection (see 4.4).

b. Verification of qualification (see 4.5).

c. Quality conformance inspection (see 4.6).

4.3 Inspection conditions and precautions.

4.3.1 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202.

4.3.2 Precautions. Adequate precautions shall be taken during inspection to prevent condensation of moisture on resistors, except during the moisture-resistance temperature-cycling tests. Precautions shall also be taken to prevent damage by heat when soldering resistor leads to terminals.

4.4 Qualification inspection. Initial qualification shall be based upon the manufacturers qualification to a minimum 'S' failure rate for each style, of the equivalent MIL-R-39007 resistor being offered for listing on the qualified products list of this specification. Additionally, the manufacturer shall pass the qualifying activity audit (see appendix A).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:46Z
Check the source to verify that this is the current version before use.

4.5 Verification of qualification. Every 6 months the manufacturer shall compile a summary of the results of quality conformance inspections in the form of a verification of qualification report, and forward it to the qualifying activity within 30 days after the end of the reporting period as the basis for continued qualification. If the summary of test results indicates noncompliance with specification requirements, and a corrective action acceptable to the qualifying activity has not been undertaken, the qualifying activity may take action to remove the failing product from the qualified products list. In addition, the manufacturer shall immediately notify within 15 days the qualifying activity whenever the data indicates that the manufacturer has failed to maintain his qualified product. Continuation shall be based on evidence that over the 6-month period the following has been met:

a. The manufacturer has not modified the design of the item.

b. Parts manufactured have met all specification requirements.

c. Lot rejection for group A inspection does not exceed 3 percent, or one lot, whichever is greater.

4.6 Quality conformance inspection. Parts supplied to this specification shall have been manufactured on the same production line as the equivalent qualified MIL-R-39007 resistor.

4.6.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspections.

4.6.1.1 Production lot (group A, subgroups 1, 1a, and 2). A production lot shall be of one design as specified in the baseline documentation. As a minimum requirement, the lot shall consist of resistors of the same style, characteristic, resistance value, voltage rating and tolerance, and protective enclosure material of the same production line using the same methods and processes defined in the baseline documentation. All resistors in the lot shall have been started, processed, assembled and tested as a group on the same manufacturing/test equipment, to the same baseline/revision, and identified with the same lot date code designation. Traceability of all raw materials used in the lot shall be maintained. The following additional restrictions shall apply:

a. The cores or encapsulation material for each lot shall be from the same raw materials lot.

b. The raw materials used in the wire winding process shall be from the same raw material lots.

4.6.1.1.1 Rework. Rework is not allowed on any operation except marking and lead rework.

4.6.1.2 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table VIII, and shall be made on the same set of sample units, in the order shown.

4.6.1.2.1 Subgroup 1. Subgroup 1 test shall be performed on a production lot basis on 100 percent of the product supplied under this specification. Resistors that are out of resistance tolerance, or which experience a change in resistance greater than that permitted for the tests of this subgroup shall be removed from the lot. Only lots having not more than 3-percent rejects or one resistor, whichever is greater due to exceeding the specified resistance change limit and arcing during DWV, as a result of subgroup 1 tests shall be furnished on orders.

4.6.1.2.1.1 Subgroup 1a. Subgroup 1a test shall be performed on a production lot basis on 100 percent of the product supplied under this specification. Resistors that are found not in compliance with the inspection criteria in appendix C shall be removed from the lot. Only lots having not more than 5-percent rejects or one resistor, whichever is greater due to noncompliance with the inspection criteria found in appendix C, as a result of subgroup 1a tests shall be furnished on orders. Corrective action shall be taken on such rejects.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:46Z
Check the source to verify that this is the current version before use.

TABLE VIII. Group A inspection.

Inspection	Requirement paragraph	Method paragraph	Sampling procedure
Subgroup 1 1/			
Conditioning	3.8	4.7.2	100 percent inspection
Thermal shock	3.12	4.7.6	
Short-time overload	3.13	4.7.8	
Dielectric withstanding voltage	3.14	4.7.9	
DC resistance 2/	3.9	4.7.3	
Visual mechanical	3.1, 3.4.2, 3.4.3 3.5, 3.5.1.4, 3.23 incl, and 3.25, 3.25.1	4.7.1	
Subgroup 1a			100 percent inspection
Radiographic	3.22	4.7.16	see 4.6.1.2.1.1
Subgroup 2			
Destructive Physical Analysis (D.P.A.)	3.18	4.7.12	5 samples 0 failures
Subgroup 3			
Solderability	3.10	4.7.4	see 4.6.1.2.5

1/ At the manufacturer's option, the determination of resistance change may be by any method which is within the accuracy requirements of this specification.

2/ Resistors shall meet the specified initial resistance tolerance after being subjected to the preceding test.

4.6.1.2.2 Manufacturer's production inspection. If the manufacturer performs tests equal to or more stringent than those specified in subgroup 1, table VIII, as the final step of this production process, group A, subgroup 1 inspection may be waived and the data resulting from the manufacturer's production tests may be used instead. Authority to waive the subgroup 1 inspection shall be granted by the qualifying activity only. The following criteria must be complied with:

a. Tests conducted by the manufacturer during production shall be clearly identical to or more stringent than those specified for subgroup 1. Test conditions shall be equal to or more stringent than those specified for subgroup 1.

b. Manufacturer subjects 100 percent of the product supplied under this specification to his production tests.

c. The parameters measured and the failure criteria shall be the same or more stringent than that specified herein.

d. The lot rejection criteria is the same or more stringent than that specified herein.

e. The manufacturer shall make available all information concerning the test procedures and instrumentation used in his production tests. This data shall be provided as part of the evaluation required for MIL-STD-790, and shall be documented in the baseline. The manufacturer shall also make available to the Government all records of all detail test data resulting from production tests.

f. Once approved the manufacturer shall not change the test procedures or criteria without prior notification and concurrence by the qualifying activity and subsequent change in baseline.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:46Z
Check the source to verify that this is the current version before use.

4.6.1.2.3 Subgroup 2 tests.

4.6.1.2.3.1 Sampling plan. Subgroup 2 tests shall be performed on a production lot basis. In the event of 1 or more failures the lot shall be rejected.

4.6.1.2.3.2 Rejected lots. The rejected lot shall be segregated from new lots and those lots that have passed inspection. The rejected lot shall be 100 percent inspected for those quality characteristics found defective in the sample and any defects found removed from the lot. A new sample shall then be randomly selected in accordance with table VIII. If one or more defects are found in this second sample, the lot shall be rejected and shall not be supplied to this specification.

4.6.1.2.4 Subgroup 3.

4.6.1.2.4.1 Sampling plan. Thirteen samples shall be selected randomly from one days' production, to form an inspection lot and subjected to the subgroup 3 solderability test. The manufacturer may use electrical rejects from the subgroup 1 screening tests for all or part of the samples to be used for solderability testing. If there are one or more defects, the lot shall be considered to have failed.

4.6.1.2.4.2 Rejected lots. In the event of one or more defects, the inspection lot is rejected. The manufacturer may use one of the following options to rework the lot:

a. Each production lot that was used to form the failed inspection lot shall be individually submitted to the solderability test as required in 4.6.1.2.5. Production lots that pass the solderability test are available for shipment. Production lots failing the solderability test can be reworked only if submitted to the solder dip procedure in b.

b. The manufacturer submits the failed lot to a 100 percent solder dip using an approved solder dip process per 3.5.1.6. Following the solder dip the electrical measurements required in group A Subgroup 1 tests shall be repeated on 100 percent of the lot. The Percent Defective Allowable (PDA) for the electrical measurements shall be as for the subgroup 1 tests. Thirteen additional samples shall be then selected and subjected to the solderability test with zero defects allowed. If the lot fails this solderability test, the lot shall be considered rejected and shall not be furnished against the requirements of this specification.

4.6.1.3 Group B inspection. Group B inspection shall consist of the tests specified in table IX, in the order shown, and the sample shall be selected from production lots that have passed group A inspection, and which are manufactured during a one week period. The group B samples shall represent all styles produced and be limited to one weeks production.

4.6.1.3.1 Sampling plan. A random sample shall be selected in accordance with table IX, if one or more defects are found, the lot shall be rescreened and defects removed. A new sample of parts shall then be randomly selected, and if one or more defects are found in the second sample, the lot shall be rejected and shall not be supplied to this specification.

TABLE IX. Group B inspection.

Inspection	Requirement paragraph	Method paragraph	Number of samples
Subgroup 1 Resistance to solvents	3.11	4.7.5	8 samples 0 failures
Subgroup 2 Insulation resistance	3.15	4.7.10	13 samples 0 failures

4.6.1.3.2 Disposition of sample units. Sample units which have been subjected to group B inspection shall not be delivered on the contract or purchase order.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:46Z
Check the source to verify that this is the current version before use.

TABLE X. Group C inspection.

Inspection	Requirement paragraph	Method paragraph	Number of sample	Number of defects allowed
Subgroup 1 Thermal shock (100 cycles)	3.12	4.7.6	10 (5 high, 5 low)	0
Subgroup 2 Moisture resistance	3.16	4.7.11	10 (5 high, 5 low)	0

4.6.2 Periodic inspection. Periodic inspection shall consist of group C inspection (see table X). Except where the results of these inspections show noncompliance with the applicable requirements (see 4.6.2.1.3), delivery of products which have passed groups A and B inspection, shall not be delayed pending the results of group C inspection. When there is noncompliance a separate sample shall be selected from lots as defined in 4.6.1.1 for each enclosure.

4.6.2.1 Sampling plan.

4.6.2.1.1 Monthly subgroups 1 and 2. Each month the specified number of sample units shall be subjected to the examination and test of table X. The sample shall be selected from a lot as defined in 4.6.1.1, and where possible shall be representative of the styles included in the lot. Over a 1-month period, the manufacturer should select samples so that a maximum variety of styles produced are tested. A separate set of sample shall be tested for each enclosure material each month.

4.6.2.1.2 Disposition of sample units. Sample units which have been subjected to group C inspection shall not be delivered on the contract or purchase order.

4.6.2.1.3 Noncompliance. If a sample fails to pass group C inspection, the manufacturer shall immediately notify the qualifying activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units or product which can be corrected and which were manufactured under essentially the same conditions with essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the qualifying activity, has been taken. After corrective action has been taken, group C inspection shall be repeated on additional sample units (all inspection, or the inspection which the original sample failed, at the option of the qualifying activity). Groups A and B inspections may be reinstituted; however, final acceptance and shipment shall be withheld until the group C reinspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure and corrective action shall be furnished to the cognizant inspection activity and the qualifying activity.

4.6.3 Inspection of packaging. The sample and inspection of preservation, packaging, and marking shall be in accordance with the requirements of MIL-R-39032.

4.6.4 Retention of data and test samples. The manufacturer shall retain the following data and test samples for a minimum of 12 years.

a. Groups B and C samples.

b. DPA samples.

c. Lot travelers.

d. Lot test information.

e. Results of special screens, corrective actions, failure analysis reports (if any).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:46Z
Check the source to verify that this is the current version before use.

4.7 Methods of inspections.

4.7.1 Visual and mechanical inspections. Resistors shall be examined to verify that the materials, design, construction, physical dimensions, marking and workmanship are in accordance with the applicable requirements (see 3.1, 3.5.1.2 to 3.5.1.7 inclusive, and 3.25 to 3.28.3, 3.29 to 3.29.1 inclusive).

4.7.2 Conditioning (see 3.8). Resistors shall be conditioned in accordance with method 108 of MIL-STD-202. The following details and exceptions shall apply:

a. Method of mounting. Supported by their terminal leads at a point .375 ±.062 from the resistor body. Resistors shall be so arranged that the temperature of any one resistor shall not appreciably influence the temperature of any other resistor. There shall be no undue draft on the resistors. If forced-air circulation is employed, the air velocity shall not exceed 500 feet per minute and there shall be no direct impingement of the forced-air supply upon the resistors.

b. Temperature and tolerance 25°C, +15°C, -0°C.

c. Initial measurements: Initial resistance shall be measured as specified in 4.7.3 at 25°C +15°C, -0°C. This initial measurement shall be used as the reference for all subsequent measurements.

d. Operating conditions: Rated dc continuous working voltage or rated working voltage from an ac power supply at commercial line frequency, intermittently, 1.5 hours on and 0.5 hour off for 100 +16, -4 hours. Where the resulting wave form is other than that of a commercial line, voltages shall be set using a "true rms" voltmeter and the peak voltage shall not exceed 2.5 times the rated continuous working voltage. Each resistor shall dissipate a wattage equal to the wattage of the resistor.

e. Measurement after condition: Resistance shall be measured at the end of 100 +16, -4 hours as specified in 4.7.2c and 4.7.3. The same procedure for measurement shall be used for initial and final measurements. Leads may be cleaned prior to final measurement.

f. Examination after conditioning: Resistors shall be examined for evidence of mechanical damage.

4.7.3 DC resistance (see 3.9). Resistors shall be tested in accordance with method 303 of MIL-STD-202. The following details and exceptions shall apply:

a. Measuring apparatus: Different types of measuring test equipment (multimeters, bridges, or equivalent) are permitted to be used on the initial and final readings of this test, provided the equipment is the same style, model or if it can be shown that the performance of the equipment is equivalent or better.

b. Limit of error of measuring apparatus: One fourth of the specified initial resistance tolerance or 0.1 percent, whichever is less, +0.002 ohm. The same measuring apparatus shall be used for any one test, but not necessarily for all tests.

c. Test voltage: Measurements of resistance shall be made by using a dc potential resulting in not more than 1 percent of rated wattage. This same voltage shall be used whenever a subsequent resistance measurement is made. For values less than 10 ohms, use a measuring device limited to 100 milliamperes or less.

d. Points of application of test voltage for initial resistance tolerance measurement: Resistors shall be measured by attaching the test leads as close to the resistor body as practical. For resistors 20 ohms and less, the referee point for attaching the test leads shall be .375 ±.062 inch from the body.

4.7.4 Solderability (when applicable) (see 3.10). Resistors shall be tested in accordance with method 208 of MIL-STD-202. The following details shall apply:

a. Both leads shall be tested.

b. The leads shall be dipped within .062 inch (1.57 mm) of the body.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:46Z
Check the source to verify that this is the current version before use.

4.7.5 Resistance to solvents (see 3.11). Resistors shall be tested in accordance with method 215 of MIL-STD-202. The following details and exceptions shall apply:

a. The marked portion of the resistor body shall be brushed.

b. The number of sample units shall be as specified in table IX, as applicable.

c. Resistors shall be examined for mechanical damage and legibility of markings.

4.7.6 Thermal shock (see 3.12). Resistors shall be tested in accordance with method 107 of MIL-STD-202. The following details and exceptions shall apply:

a. Mounting: In such a manner that there is at least 1 inch of free air space around each resistor, and in such a position with respect to the air stream that the mounting offers substantially no obstruction to the flow of air across and around the resistors. At the option of the manufacturer the units may be mounted in an equivalent manner as that specified in MIL-STD-202, method 107, as long as the recovery time is in accordance with that specified in method 107.

b. Measurement before cycling: DC resistance shall be measured as specified in 4.7.3.

c. Test condition B for group A and group C but group C is at 100 cycles.

d. Measurement after cycling: Not less than 1 hour, but within a 24 hour period after the last cycle, dc resistance shall be measured as specified in 4.7.3.

e. Examination after test: resistors shall be examined for evidence of mechanical damage.

4.7.8 Short-time overload (see 3.13). DC resistance shall be measured as specified in 4.7.3. The resistors shall then be mounted by means other than soldering and shall be subjected to an overload voltage which will result in the specified multiple of rated wattage for a 5 second duration (see 3.1). DC resistance shall again be measured after the resistors have cooled to room temperature.

4.7.9 Dielectric withstanding voltage (3.14).

4.7.9.1 Atmospheric pressure. Resistors shall be tested in accordance with method 301 of MIL-STD-202. The following details and exceptions shall apply:

a. Special preparations: Resistors shall be placed in a conductive material which will conform to the resistor surface so that between 90 and 100 percent of the outer periphery is contacted. The conductive material shall be centered on the resistor body. Care should be taken that any part of the resistor lead is as far away from the conductive material as possible.

b. Magnitude of test voltage: As specified (see 3.1).

c. Nature of potential: An alternating current (ac) supply at commercial line frequency and waveform.

d. Duration of application of test voltage: Five seconds for group A inspection.

e. Rate of application of voltage: The test voltage shall be raised from zero to the application value, as uniformly as practicable, at the following rate: 500 volts rms per second minimum for group A testing.

f. Points of application of test voltage: Between the resistor terminals connected together and mounting hardware, or the conductive material, as applicable.

g. Measurement during the test: The leakage current shall be monitored during the application of test voltage.

h. Examinations after test: Resistors shall be examined for evidence of flashover, mechanical damage, arcing, and insulation breakdown.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:46Z
Check the source to verify that this is the current version before use.

4.7.10 Insulation resistance (see 3.15). Test shall be made in accordance with method 302 of MIL-STD-202 with the following exceptions:

a. Test condition A.

b. Special preparation: In accordance with 4.7.9.1a.

c. Points of measurement: In accordance with 4.7.9.1g.

4.7.11 Moisture resistance (see 3.16). Resistors shall be tested in accordance with method 106 of MIL-STD-202. The following details and exceptions shall apply:

a. Mounting: Soldered by their leads to rigid mounts or terminal lugs. The spacing of the mounts or terminal lugs shall be such that the length of each resistor lead is approximately .375 inch (9.53 mm) when measured from the edge of the supporting terminal to the resistor body. One-half of the sample units shall be covered with a V shaped metal strap whose width is equal to the length of the resistor body as indicated on figure 2. The strap shall be made of a corrosion resistant metal and shall be kept in contact with the resistor body by supporting the body as indicated on figure 2, with a conducting, noncorrosive support whose width is less than that of the body and which will not act as a moisture trap. For group C inspection, each half of the sample shall be apportioned as three highest values, two lowest values for a total of five sample units per half. The mounting straps may be individual for each resistor or continuous for all resistors.

b. Initial measurement: Immediately following the initial drying period, dc resistance shall measured as specified in 4.7.3 at test conditions specified in 4.3.1.

c. Polarization and loading voltage:

(1) Polarization voltage: During steps 1 to 6 inclusive, a 100 volt dc potential shall be applied only to those resistors which have a polarization strap. This potential shall be applied with the positive lead connected to the resistor terminals tied together, and the negative lead connected to the polarization straps.

(2) Loading voltage: During the first 2 hours of steps 1 and 4, a dc test potential equivalent to 100 percent rated wattage shall be applied to those resistors which do not have the polarizing strap specified in 4.7.11a.

d. Subcycle: Step 7b shall not be applicable. Step 7a shall be performed during any five of the first nine cycles only.

e. Final measurements: Upon completion of step 6 of the final cycle, the resistors shall be held at the high humidity condition and a temperature of 25°C ±2°C for a period of 1.5 to 3.5 hours. Resistors shall then be removed from the chamber, and within 0.5 hour, the dielectric withstanding voltage (atmospheric), insulation resistance, and dc resistance tests shall be performed as specified in 4.7.9.1, 4.7.10, and 4.7.3, respectively. The same straps used for polarizing the resistors may also be used for the dielectric withstanding voltage and insulation resistance tests.

f. Examination after test: Resistors shall be examined for evidence of mechanical damage.

4.7.12 Destructive physical analysis (see 3.18). Resistors shall be examined as specified in appendix B of this document.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:46Z
Check the source to verify that this is the current version before use.

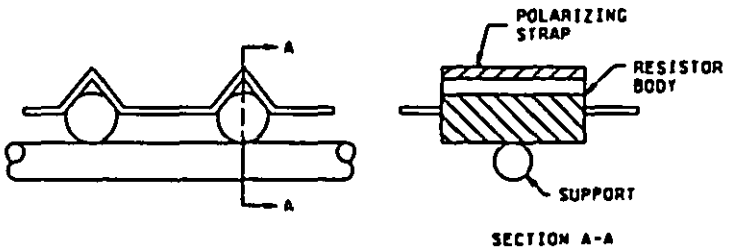


FIGURE 2. Mounting straps for moisture resistance test.

4.7.13 Outgassing (see 3.19). Resistors shall be tested in accordance with ASTM E-595. At the manufacturer's option, a certificate of compliance can be substituted for this procedure. The following details and exceptions shall apply. Samples of any materials used shall be cut into pieces having .38 inch (9.7 mm) maximum dimension. A sample of 100 to 300 milligrams of material shall be weighed and then placed in a vacuum chamber under the following conditions:

Pressure----------10^{-6} torr or less.
Temperature of specimen----Maximum operating temperature of device.
Exposure time----------24 hours.

During this exposure, a collector plate, 9.7 inches (246 mm) in diameter shall be held at 25°C ±1°C near the sample. Immediately upon removal of the specimen and collector plate from the chamber, they shall be weighed. The Total Mass Loss (TML) is sample mass before exposure minus the sample mass after exposure divided by the initial sample mass (expressed in percent change). The Volatile Condensable Material (VCM) is the increase in the weight of the collector plate divided by the mass of sample (expressed in %).

4.7.14 Fungus (see 3.20). Unless certification is provided, resistors shall be tested in accordance with method 508 of MIL-STD-810.

4.7.15 Reactance (applicable to "N" and "Z" terminals and windings only) (see 3.21). Effective series inductance and parallel capacitance shall be determined using methods I and II, as applicable (see 4.7.15.1 and 4.7.15.2). The units shall be mounted by their normal mounting means using the fixtures as established in the following procedures.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:46Z
Check the source to verify that this is the current version before use.

4.7.15.1 Method I. This method shall be used for resistance values of 100 ohms and below. Use Boonton electronics model 63H or equivalent. Effective inductance of the resistors shall be measured at a frequency of 0.5 megahertz using the test fixtures TF-A or TF-B, as applicable and shown on figure 3. The appropriate test fixture shall be mounted to the binding post of the test bridge. The appropriate shorting bar conforming to figure 6 shall be inserted into clips of the test fixtures so that the terminals rest firmly against the stops and the shorting bar is centered between the test fixture terminals. The bridge shall then be nulled and calibrated prior to the inductance readings. After removing the shorting bar, insert the resistor to be tested so that the leads ate straight and rest firmly against the stops and the unit is centered between the test fixture terminals. The bridge shall be adjusted until the meter indicates a null. The inductance can be read directly from the bridge.

4.7.15.2 Method II. This method shall be used on resistance values above 100 ohms. Use Boonton radio RX meter model 250A or equivalent. Effective inductance of the resistors shall be measured at a frequency of 1 megahertz using the test fixtures TF-C or TF-D, as applicable, and shown on figure 4. The appropriate test fixture shall be mounted to the binding posts of the RX meter. The appropriate shorting bar conforming to figure 6 shall be inserted in the clips of the test fixtures so that the terminals rest firmly against the stops and the shorting bar is centered between the test fixture terminals. The bridge shall then be nulled and calibrated prior to inductance readings. After removing the shorting bar, insert the resistor to be tested so that the leads are straight and rest firmly against the stops and the unit is centered between the test fixture terminals. The bridge shall be adjusted until the meter indicates a null.

4.7.15.3 Cp readings. If the Cp reading falls within the:

a. Positive portion of the dial, the resistor is capacitive and the equivalent parallel capacitance is read directly from the dial.

b. Negative portion of the dial, the resistor is inductive and the equivalent series inductance can be computed using the following formulas:

$$Q = 2 * f * Rp * CP$$

$$Ls = \frac{Cp * Rp^2}{1 + Q^2}$$

Where: f = frequency in hertz
Rp = indicated parallel resistance
Cp = indicated parallel capacitance
Ls = Equivalent series inductance

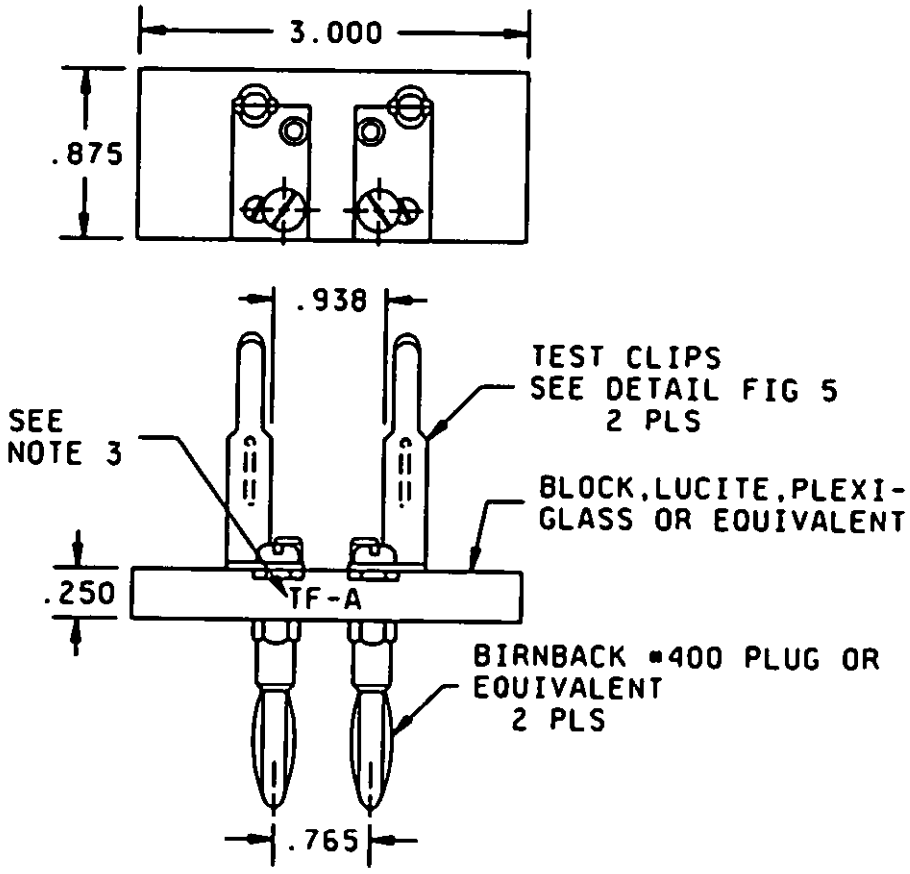
4.7.15.4 Examination after test. Units shall not exceed the limits specified in 3.21.

4.7.16 Radiographic inspection (3.22). In accordance with appendix C; two views 90 degrees apart by X-ray, or 360 degrees view by Vidicon. Use of "real-time" X-ray systems capable of viewing through 360 degrees of rotation is encouraged.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-R-39032.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:46Z
Check the source to verify that this is the currentversion before use.



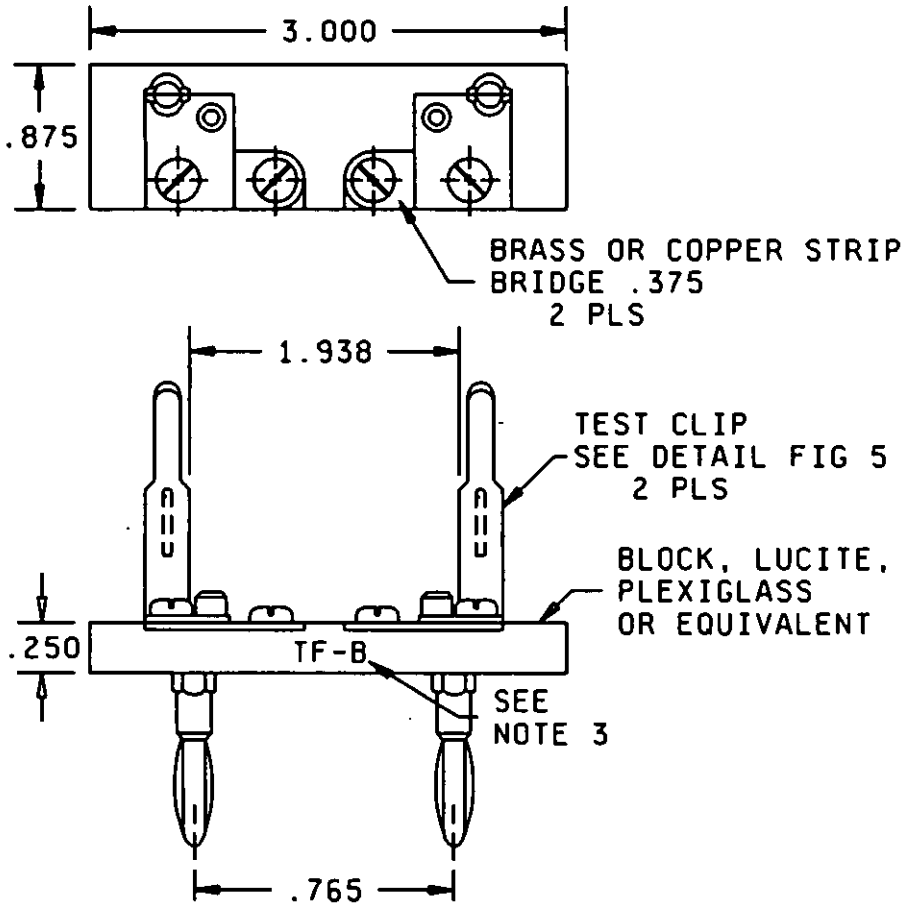
TF-A FIXTURE

Inches	mm
.005	0.13
.250	6.35
.765	19.43
.875	22.23
.900	22.86
.938	23.83
3.000	76.20

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Identification to be marked on fixture.
4. Tolerance is ±.005 (0.13 mm).
5. TF-A fixture shall be used when resistor body length is .900 (22.86 mm) or less.

FIGURE 3. Test fixture for method 1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:46Z
Check the source to verify that this is the current version before use.



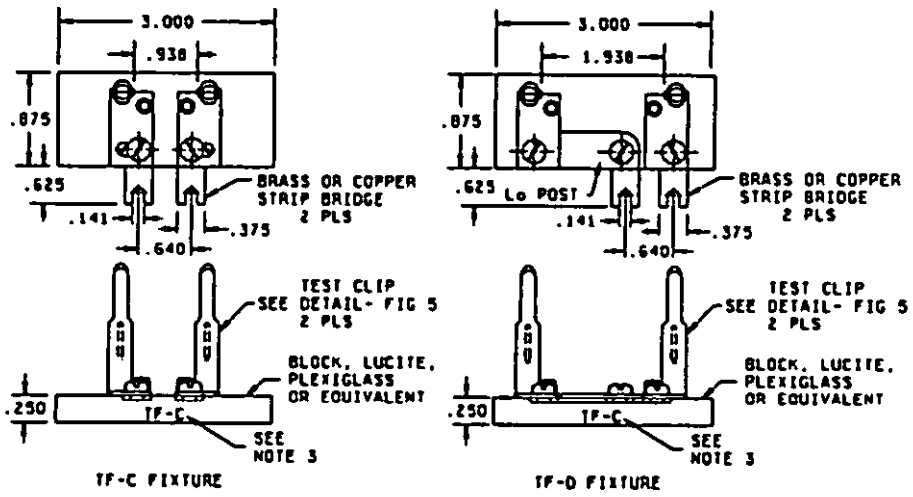
TF-B FIXTURE

Inches	mm
.005	0.13
.250	6.35
.375	9.53
.765	19.43
.875	22.23
.900	22.86
1.938	49.23
3.000	76.20

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Identification to be marked on fixture.
4. Tolerance is ±.005 (0.13 mm).
5. TF-B fixture shall be used when resistor body length is .900 (22.86 mm) or less.

FIGURE 3. Test fixture for method I - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:46Z
Check the source to verify that this is the current version before use.

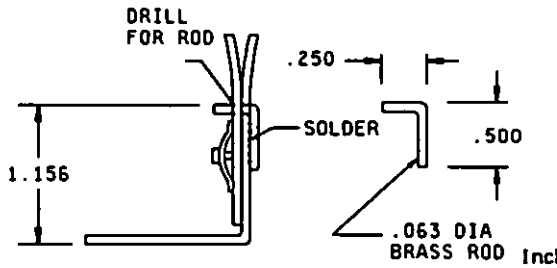


Inches	mm
.005	0.13
.141	3.58
.250	6.35
.375	9.53
.625	15.88
.640	16.26
.875	22.23
.900	22.86
.938	23.83
1.938	49.23
3.000	76.20

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Identification to be marked on fixture.
4. Tolerance is ±.005 (0.13 mm).
5. TF-C fixture shall be used when resistor body length is .900 (22.86 mm) or less.
6. TF-D fixture shall be used when resistor body length is greater than .900 (22.86 mm).

FIGURE 4. Test fixtures for method 11.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:46Z
Check the source to verify that this is the current version before use.



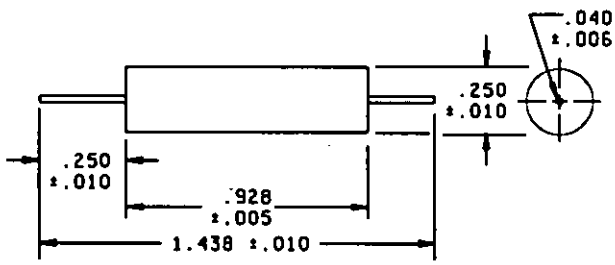
"RAPID TEST" CLIP R OR L OR EQUAL

DETAIL

Inches	mm
.063	1.60
.250	6.35
.500	12.70
1.156	29.36

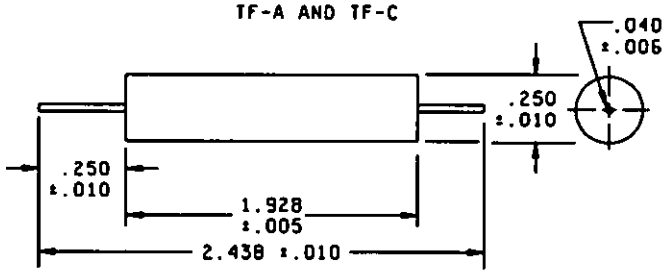
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Tolerance ±.005 inch (0.38 mm).

FIGURE 5. Test clip detail.



COPPER OR BRASS

TF-A AND TF-C



COPPER OR BRASS

TF-B AND TF-C

Inches	mm
.005	0.13
.006	0.15
.010	0.25
.040	1.02
.250	6.35
.928	23.57
1.438	36.53
1.928	48.97
2.438	61.93

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 6. Shorting bar for test fixtures.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:46Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Ordering data.

6.1.1 Acquisition requirement. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Title, number, and date of the applicable detail specification, and the PIN (see 1.2.1 and 3.1).

c. Levels of preservation, packaging, and applicable marking (see section 5)

6.2 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening bids, qualified for inclusion in the applicable qualified products list whether or not such products have actually been so listed by that date. The attention of the manufacturers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Air Force Space Systems and Missile Center; however, information pertaining to qualification of products may be obtained from Defense Electronics Supply Center (DESC-E), 1507 Wilmington Pike, Dayton, OH 45444-5278.

6.3 Derating. The intention of this specification is to cover resistors capable of full load operation with a high degree of stability at any ambient temperature up to including 25°C. However, if it is desired to operate these resistors at ambient temperatures greater than 25°C, the resistors should be derated in accordance with figure 1.

6.4 Mounting. Under conditions of severe shock or vibration, or a combination of both, resistors of all sizes described in this specification should be mounted so that the body of the resistor is restrained from movement with respect to the mounting base. It should be noted that if clamps are used, certain electrical characteristics of the resistor will be altered. The heat dissipating qualities of the resistor will be enhanced or retarded depending on whether the clamping material is a good or poor heat conductor. Under less severe vibration conditions, all styles may be supported by their leads only. The lead lengths should be kept as short as possible, .250 inch or less preferred, but not longer than .625 inch (15.88 mm). The longer the lead, the more likely that a mechanical failure will occur.

6.5 High frequency. Resistors should not be used in circuits where their ac performance is of critical importance in the operation of such circuits.

6.6 Coating materials. Certain coating materials used in fabricating resistors to this specification may be subject to "outgassing" of volatile material when operated at surface temperatures over 200°C. This phenomena should be taken into consideration for equipment design.

6.6.1 Tin plated finishes. Tin plating is prohibited (see 3.4.7) since it may result in tin whisker growth. Tin whisker growth could adversely affect the operation of electronic equipment systems. For additional information on this matter refer to ASTM B545 (Standard Specification for Electrodeposited Coating of Tin).

6.7 Maintenance and replacement of lead types. For maintenance and replacement or other logistical purposes, solderable leads (S) shall be used to support weldable leads (W). Weldable leads should not be stocked.

6.8 Reactance resistors. When low reactance resistors are required with values below 10 ohms, the S and W windings can be used since these values have low reactance and meet the requirements specified in table VI.

6.9 Caution in handling RWS resistors. Dust from the materials of construction of these resistors are potentially hazardous if inhaled or ingested. Do not machine, abrade, file, ingest, etc. without proper safety equipment and procedures.

6.10 Performance characteristics. These are for reference only, see table XVII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:46Z
Check the source to verify that this is the current version before use.

6.11 Subject term (key word) listing.

Solderable
Weldable
Inductively wound
Noninductively wound

6.12 Part or Identifying Number (PIN). PIN is a new term encompassing previous terms used in specifications such as part number, type designator, identification number etc. (see 1.2.1).

TABLE XVII. Performance characteristics.

Style		RWS81	RWS84	RWS89
Maximum resistance temperature characteristic in parts/million /°C (ref. to 25°C)	0.1 ohm to 0.499 ohm	+650 PPM	+650 PPM	+650 PPM
	0.499 ohm to 1.0 ohm	+400 PPM	+400 PPM	+400 PPM
	1.0 ohm to 10 ohms	±50 PPM	±50 PPM	±50 PPM
	10 ohms and above	±20 PPM	±20 PPM	±20 PPM
Maximum ambient temperature at rated wattage derating (see figure 1)		25°C	25°C	25°C
Maximum ambient temperature at zero wattage (see figure 1)		250°C	250°C	250°C
Power rating in watts		1	7	3
Conditioning (see 3.8)		.2	.2	.2
Thermal shock (see 3.12)		.2	.2	.2
Low temperature storage 2/		--	--	--
Short time overload (see 3.13)		.2	.2	.2
Dielectric withstanding voltage (see 3.14)		.1	.1	.1
Moisture resistance (see 3.16)		.2	.2	.2
Shock, specified pulse 2/		--	--	--
Vibration, high frequency 2/		--	--	--
Life 2/		--	--	--
High temperature exposure 2/		--	--	--
Insulation resistance (see 3.15)		1,000 megohms minimum	1,000 megohms minimum	1,000 megohms minimum
Insulation resistance, after moisture resistance (see 3.16)		100 megohms minimum	100 megohms minimum	100 megohms minimum
Resistance tolerance ± percent (see table IV)		B, D, F	B, D, F	B, D, F

1/ Where total resistance change is 1 percent or less, but shall be considered as ±(___ percent +0.005 ohm).

2/ As describe in scope, these tests shall be cable of meeting the requirements of MIL-R-39007.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:46Z
Check the source to verify that this is the current version before use.

APPENDIX A

PROCEDURES FOR THE PRODUCT AND QUALITY AUDIT

10. SCOPE

10.1 Scope. The appendix provides procedures to the Product and Quality Audit team for the auditing of manufacturers supplying to this specification for certification of manufacturing facilities. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. INSTRUCTIONS

30.1 Baseline document. The manufacturer shall complete the baseline document and submit the required information to the qualifying activity prior to scheduling the Product and Quality Audit. It is not intended that the manufacturer submit proprietary information. The baseline document shall contain the following information, as a minimum:

a. Detailed flow charts indicating the following:

(1) All significant process steps and measurement and inspection points by descriptive name and document number.

(2) All raw material and interim and finished part quality control points by descriptive name, document number, and location in the processing sequence. Documents shall include forms and travelers.

(3) All raw material incoming inspection, storage, and processing steps by descriptive name, document number, and location in the processing sequence. Documents shall include forms and travelers.

(4) The identification of all critical (significant to end product properties) manufacturing and inspection steps as determined by the manufacturer in conjunction with the audit team at the time of baseline document approval.

b. A listing of the documents identified in the flow charts identifying current revision letter, revision date and classification (i.e., critical, proprietary).

c. Blank copies of all current forms and travelers cited in the flow charts.

d. Copies of quality control documents detailing sampling plans used in inspections cited on flow charts and the quality control inspector training.

e. Procedures to control documentation, documentation changes, and distribution of the documents or document changes to all required personnel.

f. Procedures for raw material control for procurement, incoming inspection, storage, withdrawal from inventory, and shelf life control.

g. Details of part design delineating dimensions, tolerance, materials,and internal part numbers.

h. Procedures for controlling computer software including basic flow chart for program logic, current revision numbers, program access controls, and storage of part revisions of programs.

i. A complete list of process and test equipment utilized.

j. Organizations chart of key personnel and their relations.

k. List of raw materials and approved/qualified supplies.

l. List of approved outside test and service laboratories used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:46Z
Check the source to verify that this is the current version before use.

m. Procedures for statistical process control and failure analysis.

30.1.1 Changes to baseline. The audit team shall compare the latest baseline document to the previous baseline document to assure all changes have been submitted to the qualifying activity and have been approved.

a. Minor changes. Any subsequent departure from the approved flow chart (i.e., additions or deletions) or changes to the revision status of those processes or inspections are allowed, provided they do not violate the changes prohibited by section b (major changes). These minor changes shall be described in general terms to the audit team during subsequent audits. This action shall result in an update of the baseline document.

b. Major changes. No changes (i.e., substitutions, additions, deletions, or modifications) shall be allowed in the following areas without prior review and approval of the qualifying activity. In these, and other areas deemed major by the audit team at the time of baseline approval, partial or total product requalification may be required as determined by the qualifying activity.

(1) Significant raw materials.

(2) Qualified sources for significant raw materials.

(3) Substrate preparation.

(4) Procedure for winding.

(5) Pitch of winding.

(6) Capping and lead attachment.

(7) End cap plating and cleaning.

(8) Coating process.

(9) Encapsulation or sealing process.

(10) Group A, B, or C inspections.

30.2 Audits. An initial audit and subsequent periodic audits (every year or as deemed necessary) shall be performed by the qualifying activity. The audit shall include, but not be limited to, all of the items on the checklist (plus MIL-STD-790) and any GIDEP ALERTS issued against the manufacturer at the facility of interest. The results of any additional items that are assessed shall be added to the checklist for record purposes.

30.2.1 Duties of the audit team.

a. The audit team shall review the manufacturer's facility and the proposed baseline documentation for adequacy, accuracy and completeness during the audit of the manufacturer. The proposed baseline documentation should be used as a baseline/reference during the course of the audit.

b. The audit team shall verify that the test measuring procedures and equipment are sufficiently accurate for the required test.

c. The audit team shall typically sample check twelve months of records for the items being certified. In addition, the location of the equipment and the thoroughness and adequacy of the operating procedures and instructions shall be checked. Whenever practical, operations utilizing critical equipment and procedures shall be witnessed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:46Z
Check the source to verify that this is the current version before use.

APPENDIX A

TABLE XVIII. Check list.

Check - item	Inspection	Satisfactory
(a) Verification of MIL-STD-790 requirements.		
(b) Flow chart (1) Key processes identified with revision. (2) Key inspection points identified with revision. (3) Quality control points identified with revision. (4) Key raw material control documents identified with revisions. (5) Critical steps identified. (6) Nonproprietary documents reviewed. (7) Test and measurements points identified with revision.		
(c) Baseline: Shall describe, as a minimum, the following: (1) Resistor design by part number indicating the following with the necessary tolerance defined: End cap alignment Body length Body diameter Shell end metallization Wire size (2) Documents listing suppliers qualified for all raw materials.		
(d) Lot control plan (see 4.6.1.1).		
(e) Traceability plan.		
(f) Training plan for all personnel.		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:46Z
Check the source to verify that this is the current version before use.

APPENDIX B

PROCEDURES FOR DESTRUCTIVE PHYSICAL ANALYSIS (DPA)

10. SCOPE

10.1 Scope. This appendix provides the methods and detailed requirements for DPA of wire-wound resistors. This appendix is a mandatory part of the specification contained herein and is intended for compliance.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. EQUIPMENT AND PROCEDURES

30.1 Equipment.

30.1.1 Sample preparation. A mechanical or chemical process shall be use to remove the outer insulating materials. This process shall cause minimal damage to the internal resistor structure. When resistors contain beryllium-oxide cores, there shall be no machining, grinding, filling or polishing performed on the cores. Beryllium-oxide dust is highly toxic.

30.1.2 Optical equipment. A binocular microscope with a minimum capability of 100x magnification and an integral light source or fiber optic light ring shall be used.

30.2 Resistor, fixed, wire-wound, style RWS81, 84, and 89.

30.2.1 Method.

30.2.2 External visual. Examine resistors at 20x minimum magnification, and examine terminals, leads, marking, general dimensions, and appearance for any evidence of defective workmanship.

40. SAMPLE PREPARATION

40.1 Conformably coated resistors. Conformal coating can be mechanically or chemically removed. When mechanical means are employed, care must be taken not to induce any rejectable defect on the resistor itself (see 30.1.1 when core is beryllium-oxide). When chemically stripping the coating, use a process and material which dissolves the coating and exposes, but not attacks, the wire, core, end terminations, leads, and welded connections, and which does not discolor or stain any surface. As required, perform SEM inspection and check for corrosion.

50. DATA RECORDS

50.1 Data records. DPA findings that deviate from configuration and other requirements shall be documented as defects.

50.2 Evaluation criteria. When the DPA is conducted as a lot conformance test, the production lot shall be rejected if one or more of the DPA samples exhibit any of the defects listed below. All exposed inner surfaces of each resistor shall be examined at 30x minimum magnification for the following characteristics:

a. End cap misalignment greater than 5 degrees.

b. End cap showing corrosion, lifted or missing plating.

c. Weld splatter at lead-to-end cap termination, or cracks in weld joint.

d. Cracked, split, or holes (from welding operation) on end caps.

e. Sudden kinks, bends, or sharp distortion on the resistance wire that reduces the wire diameter to 5/6ths or less of the initial value.

f. Loose windings on active portion of the resistor.

g. Wire not secure at weld on end cap.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:46Z
Check the source to verify that this is the current version before use.

h. Number of wire turns different between samples having similar resistance values, up to 5 percent or one turn, whichever is greater.

i. Space between wire turns more than five times the wire diameter except for values less than 1 ohm, or space between turns of less than the wire diameter except for high-value resistors using insulated wires.

j. Cracks, spalls or surface holes on the core that exceed .025 inch (0.06 mm) in the greatest dimension.

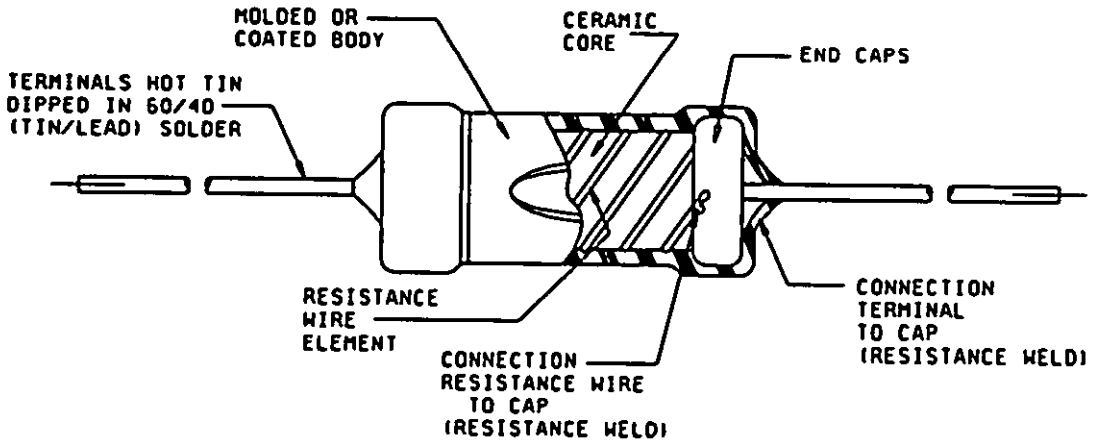


FIGURE 7. Typical wire-wound resistor, RWS style.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:46Z
Check the source to verify that this is the current version before use.

APPENDIX C

RADIOGRAPHIC INSPECTION

10. SCOPE

10.1 Scope. This appendix specifies procedures and practices necessary for radiographic inspection of wire-wound resistors during group A inspection. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance only.

10.2 Facilities. Radiographs of the resistors may be made by the resistor manufacturer or a suitable independent laboratory. In either case, facilities, equipment, personnel, and techniques shall be in accordance with the requirements specified herein.

10.3 Responsibility for inspection. The resistor manufacturer shall perform the inspection of the radiographic films and acceptance/rejection of the resistors.

20. APPLICABLE DOCUMENTS. This section is not applicable to this document.

30. EQUIPMENT

30.1 Dimensions. The dimensional values upon which these requirements are based were developed using the U.S. system. The metric equivalents are obtained by direct conversion based upon 1 inch = 25.4 mm. Where such equivalents do not correspond to a standard metric size, the closest standardized metric equivalent size may be used.

30.2 Radiography. The X-ray equipment shall have sufficient voltage range to produce radiographs in accordance with this document. The equipment shall have a focal spot of .14 inch (3.5 mm) or less and shall maintain a sharply defined image at a focal film distance of 30 to 60 inches (76.20 to 152.40 centimeters).

30.3 Exposure factors. The X-ray exposure factors shall be selected to achieve maximum image detail within the sensitivity requirements. The film shall be exposed in accordance with the following requirements:

a. X-ray voltage - lowest voltage possible.

b. H and D film density - 1.5 to 3.0.

c. Milliampere and time settings - adjusted, as necessary, to obtain satisfactory exposure.

30.4 Film. The X-ray film shall be single emulsion and of a grade defined as very fine grain.

30.4.1 Sensitivity. X-ray film and equipment shall be capable of detecting metallic particles with a major dimension of .004 inch (0.10 millimeters) or greater.

30.4.2 Exposure. Exposure factors such as KVP, current, and time shall be compatible with the sensitivity requirements of 30.4.1.

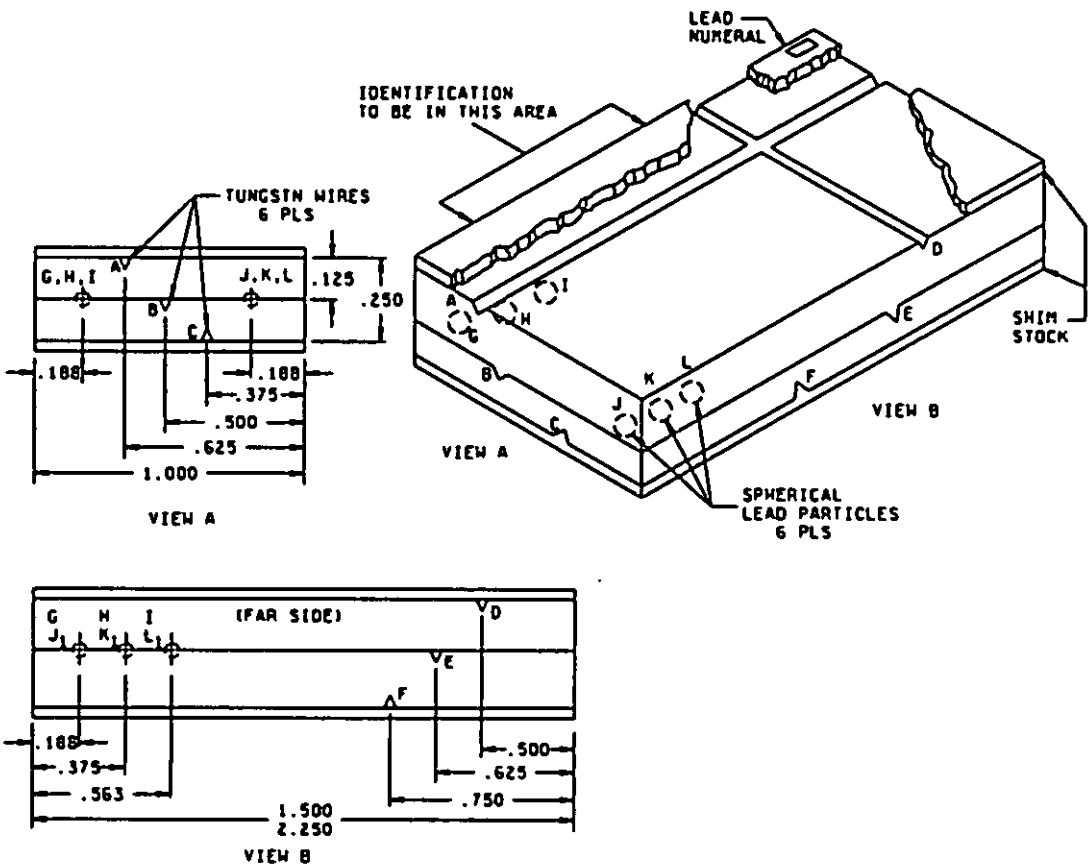
30.4.3 Film density. The X-ray equipment and processing techniques shall be capable of producing H and D film density of 1.5 to 3.0 in accordance with the American Standard Printing density type P-2.

30.4.4 Film dimensions. Radiographic film shall not exceed 14 inches (35.56 centimeters) in width and 17 inches (43.18 centimeters) in length.

30.4.5 Processing. The exposed X-ray film shall be processed in such a manner that the film shall be free of processing defects, i.e., fingerprints, chemical spots, blemishes, etc.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:46Z
Check the source to verify that this is the current version before use.

APPENDIX C



Inches	mm	Inches	mm
.125	3.18	.625	15.88
.188	4.78	.675	17.14
.250	6.35	.750	19.05
.375	9.52	1.000	25.40
.500	12.70	1.500	38.10
.563	14.30	2.250	57.15

FIGURE 8. Penetrameter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:46Z
Check the source to verify that this is the current version before use.

APPENDIX C

Tungsten wire diameters						Lead particle diameters						Steel shim stock
A	B	C	D	E	F	G	H	I	J	K	L	
.002 (.05)	.001 (.03)	.0005 (.013)	.0005 (.013)	.001 (.03)	.002 (.05)	.015 (.38)	.010 (.25)	.008 (.20)	.006 (.15)	.004 (.10)	.002 (.05)	None
|	|	|	|	|	|	|	|	|	|	|	|	.002 (.05)
|	|	|	|	|	|	|	|	|	|	|	|	.005 (.13)
| ▼	| ▼	| ▼	| ▼	| ▼	| ▼	|	|	|	|	|	|	.007 (.18)
.003 (.08)	.002 (.05)	.001 (.03)	.001 (.03)	.002 (.05)	.003 (.08)	|	|	|	|	|	|	.010 (.25)
.003 (.08)	.002 (.05)	.001 (.03)	.001 (.03)	.002 (.05)	.003 (.08)	|	|	|	|	|	|	.015 (.38)
.005 (.13)	.003 (.08)	.002 (.05)	.002 (.05)	.003 (.08)	.005 (.13)	|	|	|	|	|	|	.025 (.64)
.005 (.13)	.003 (.08)	.002 (.05)	.002 (.05)	.003 (.08)	.005 (.13)	| ▼	| ▼	| ▼	| ▼	| ▼	| ▼	.035 (.89)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Wires to be tungsten, shim stock to be carbon steel, particles to be lead. Center section to be .125 (3.18 mm) layers of clear acrylic plastic, bonded with clear plastic cement of low X-ray density. Fasteners may be used within .250 inch (6.35 mm) or less from each corner, but shall not interfere with end of use of the penetrameter. Bottom surfaces shall be flush.
4. All dimensions, shown are ±.005 inch (0.13 mm), except wires, and shim stock, which shall be within standard mill tolerances and lead particles which shall be ±.0002 (.005 mm). Groove details are not critical except that the wire must be embedded flush or below surface of plastic and centered at location shown. Particle-hole sizes are not critical, but should not exceed .031 (0.79 mm) in diameter and depth, and must be centered as shown (±.005 inch (0.13 mm)).
5. Additional layers of shim stock may be used as necessary.
6. Identification marking shall be permanent and legible. Location and size of characters is not critical but shall not interfere with or obscure the radiographic image details.

FIGURE 8. Penetrameter - Continued.

30.4.6 Film identification. Each radiographic film shall be identified with the following information:

a. Part manufacturer's name.

b. PIN (as marked on part).

c. Part cross reference.

d. Date or lot code (as marked on part).

e. View number.

f. X-ray laboratory name.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:46Z
Check the source to verify that this is the current version before use.

g. Penetrameter image.

h. Penetrameter number (see figure 8).

30.5 Penetrameters. Penetrameters shall be employed in all radiographic testing and shall be as specified on figure 8. The penetrameter image shall be used to determine radiographic quality and shall meet the following requirements:

a. Penetrameter wires shall be visible on each radiograph.

b. Penetrameters shall be selected to give a film density within +10 percent of the density of the area of immediate interest.

c. Penetrameters shall be placed on diagonal corners on the source side of the film. The plane of the penetrameters shall be normal to the radiation film. The plane of the penetrameters shall be normal to the radiation beam. When 35-mm film strip is used, the penetrameter shall be placed in a position normally occupied by a part, and a penetrameter image shall be made (exposed) for every 50 parts or 17 inches (43.18 centimeters) of film, whichever is more convenient.

d. Distortion of any penetrameter shall not exceed 10 percent.

e. The spacing between wires of a penetrameter shall not be distorted by more than 10 percent. The percentage of distortion as used in this standard is defined as follows:

Percentage distortion = $[(S_0 - S_1)/S_0] \times 100$, where S_0 = actual wire spacing, and S_1 = wire spacing as it appears on the X-ray film.

30.6 Fixtures. Suitable fixtures shall be used for mounting the electronic parts during the X-ray operation.

30.6.1 Mounting of parts. The resistors may be mounted in any type of fixture, provided that any metallic portion of the fixture is not between the body of the resistor and the film.

30.6.2 Identification of resistors. Each resistor shall be traceable from its position on the fixture to its corresponding image on the film until completion of film inspection and identification of any with rejectable defects.

30.7 Inspection and examination of radiographs. Inspection of radiographs shall be conducted by the device manufacture. Each radiograph shall be examined, utilizing the equipment specified herein. The radiographs shall be inspected to determine that each capacitor conforms to the requirements of this document.

30.7.1 Viewing equipment. The radiograph shall be examined on a suitable illuminator with variable intensity, or on a viewer suitable for radiographic inspection of projection type viewing equipment.

30.7.2 Magnification. A magnification of 10 power minimum shall be used for radiographic examination.

40. REPORTS AND RECORDS

40.1 Report of inspection. Unless otherwise specified by the contractor, the testing activity shall furnish inspection reports signed by an authorized representative of the testing activity. The reports shall give the results of the radiographic inspection and shall list the part number, the number of parts inspected, the number of parts rejected, and the date of the test.

40.2 Records of inspection. A complete record of details of inspection shall be kept by the manufacturer or testing laboratory. The record shall list the voltage potentials and currents used in the radiographic process, the time of exposure, the distance of the source of radiation from the surface of the part, the distance of the film from the same surface, the approximate angle between the central beam of radiation and the film, the screens and filters used, the size of the focal spot, the time of development of the film and the PIN and lot or date code of the resistors.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:46Z
Check the source to verify that this is the current version before use.

40.3 Records of radiographs. Each radiograph shall carry a radiograph inspection serial number or code letters to identify the radiograph with the parts examined shown in the radiograph. One complete set of radiographs shall be kept by the manufacture for a period of 3 years minimum.

50. PERSONNEL

50.1 Radiographer. Personnel engaged in radiographic processing shall be familiar with the requirements of this standard and with all applicable documentation controlling radiographic quality of parts and material being inspected. They shall be certified by their employer to be capable of producing radiographs which meet the requirements of all applicable documentation.

50.2 Radiographic interpreters. Personnel engaged in the interpretation of radiographs shall be familiar with the requirements of this standard and with all applicable documentation controlling radiographic quality of parts and materials being inspected. They shall be certified by their employer to be capable of evaluating radiographs to determine conformance of parts and materials to the requirements of all applicable documentation.

50.3 Vision. The minimum vision requirements for visual acuity of personnel inspecting film shall be as follows:

a. Distant vision shall equal to 20/30 in at least one eye, either corrected or uncorrected.

b. Near vision shall be such that the individual can read Jaeger type No. 2 at a distance of 16 inches (40.64 centimeters) either corrected or uncorrected.

50.4 Vision tests. Vision tests shall be performed by an oculist, optometrist, or by other professionally recognized personnel. One year form the effective due date of original certification, and each year thereafter, certified personnel shall be required to pass the vision tests specified herein.

60. ACCEPT REJECT CRITERIA

60.1 Accept reject criteria. Any defects shown on figure 9.

60.1.1 End caps. See 3.4.4.

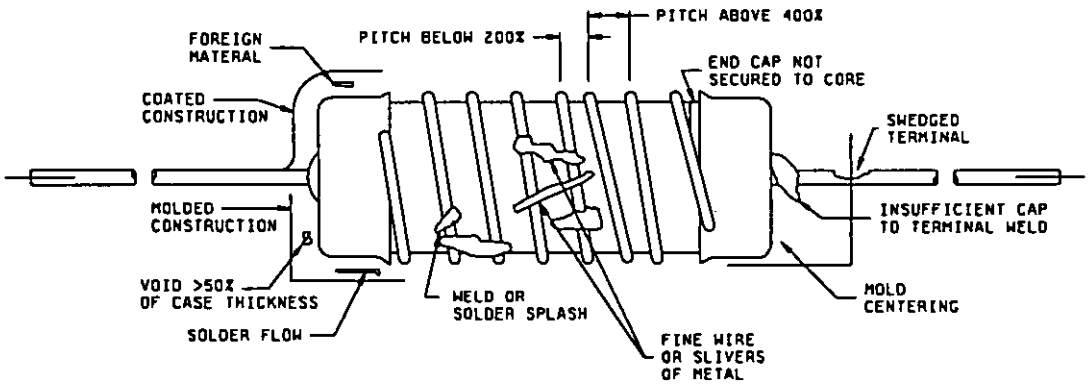


FIGURE 9. Radiography visual reject criteria.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:46Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 19
NASA - NA

Review activities:
Army - AR, AT, AV, ME
Navy - AS, CG, MC, OS
Air Force - 17, 85, 99

Preparing activity:
Air Force - 19

Agent:
DLA - ES

(Project 5905-1258)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:46Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-R-83579	2. DOCUMENT DATE (YYMMDD)

3. DOCUMENT TITLE

RESISTORS, FIXED, WIRE-WOUND (POWER TYPE), HIGH RELIABILITY, GENERAL SPECIFICATION FOR

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME SMC/SDFP P.O.C ALAN WANG	b. TELEPHONE (Include Area Code) (1) Commercial: 213-643-1966 (2) AUTOVON: 833-1966
c. ADDRESS (Include Zip Code) P.O. Box 92960 Los Angeles AFS Los Angeles, CA 9009-2960	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:46Z
Check the source to verify that this is the current version before use.